Runchu Tian

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Education

University of North Carolina at Chapel Hill, PhD in Computer Science – Chapel Hill, NC — Aug. 2026 - Present
MURGe-Lab | Advisor: Mohit Bansal

University of Illinois Urbana-Champaign, M.S. in Computer Science – Urbana, IL — Aug 2024 – May 2026
Data Mining Group | Advisor: Jiawei Han | GPA: 4.00/4.00

Tsinghua University, B.S. in Mathematics and Physics – Beijing, China — Sept 2020 – June 2024
THUNLP | Advisor: Zhiyuan Liu | GPA: 3.80/4.00

Experience

Visiting Student, University of California San Diego – San Diego, CA — June 2025 – Aug 2025
Shang Data Lab | Advisor: Jingbo Shang

- Developed SimSD, a training-free speculative decoding algorithm for diffusion language models (NeurIPS 2026).

Visiting Student, University of Washington – Seattle, WA — June 2023 – Sept 2023
Wang Lab | Advisor: Sheng Wang

- Explored hybrid gene representation combining graph and text embeddings from biological knowledge sources.

Publications

* denotes equal contribution.

2026 @ UNC-CH: *How to make post-train for long-horizon agent tasks more effective, efficient and legible*

MiRL: Error-Targeted Reinforcement Learning for Correcting Terminal Agent Misbehaviors — **Preprint**

Runchu Tian, Zaid Khan, Zun Wang, Justin Chih-Yao Chen, Mohit Bansal

Overview: We try error-targeted learning on terminal agents and find that, with < 200 envs in TMax-15K & 35% of TMax's rollout, we can beat TMax-4B full recipe on Terminal-Bench Lite (51.8% vs. 47.4%) and Terminal-Bench 2.1 (21.3% vs. 20.6%).

2025 – 2026 @ UC San Diego: *How to make the decoding of diffusion LLMs more natural*

SimSD: Simple Speculative Decoding in Diffusion Language Models — **NeurIPS 2026**

Junxia Cui*, Haotian Ye*, **Runchu Tian***, Hongcan Guo, Jinya Jiang, Haoru Li, Chaojie Ren, Yiming Huang, Kaijie Zhu, Zhongkai Yu, Kun Zhou, Jingbo Shang

Overview: A simple but effective speculative decoding algorithm for diffusion language models, which adopts a plug-and-play masking strategy that equips dLLMs with temporally valid token-level contexts for speculative decoding.

2024 – 2026 @ UIUC: *How to empower information retrieval and extraction with LLMs and semantic structures*

LLM-Based Compact Reranking with Document Features for Scientific Retrieval — **KDD 2026**

Runchu Tian*, Xueqiang Xu*, Bowen Jin, SeongKu Kang, Jiawei Han

Overview: Proposed compact document representations for LLM-based listwise reranking in scientific retrieval.

Beyond True or False: Retrieval-Augmented Hierarchical Analysis of Nuanced Claims **ACL 2025**

Priyanka Kargupta*, **Runchu Tian***, Jiawei Han

Overview: Developed ClaimSpect for decomposing nuanced claims into retrieval-grounded aspect trees.

ObSchema: From Local Observations to Global Entity Attribute Schemas in Scientific Literature **EMNLP 2026**

Xueqiang Xu, Wonbin Kweon, Jimeng Shi, **Runchu Tian**, Yunfan Kang, Wei Hu, Shaowen Wang, Jiawei Han

Overview: An observation-to-schema framework that extracts local entity-property observations, aggregates corpus-level evidence to identify recurring attributes, and refines them through value-aware merging and adaptive grouping.

PairSem: LLM-Guided Pairwise Semantic Matching for Scientific Document Retrieval **WWW 2026**

Wonbin Kweon, **Runchu Tian**, SeongKu Kang, Pengcheng Jiang, Zhiyong Lu, Jiawei Han, Hwanjo Yu

Overview: Modeled scientific concepts as entity-aspect pairs for fine-grained document retrieval.

Topic Coverage-based Demonstration Retrieval for In-Context Learning **EMNLP 2025**

Wonbin Kweon, SeongKu Kang, **Runchu Tian**, Pengcheng Jiang, Jiawei Han, Hwanjo Yu

Overview: Selected in-context demonstrations by covering fine-grained topical knowledge needs.

DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning **COLM 2025**

Pengcheng Jiang, Jiacheng Lin, Lang Cao, **Runchu Tian**, SeongKu Kang, Zifeng Wang, Jimeng Sun, Jiawei Han

Overview: Optimized LLM query rewriting for real search engines and retrievers using reinforcement learning.

Retrieval and Structuring Augmented Generation with Large Language Models **KDD 2025**

Pengcheng Jiang, Siru Ouyang, Yizhu Jiao, Ming Zhong, **Runchu Tian**, Jiawei Han

Overview: Tutorial on structured knowledge and retrieval-augmented generation with large language models.

2020 – 2024 @ THU: *Early exploration of LLM agents on tool calling, coding, and long-context capabilities*

Distance between Relevant Information Pieces Causes Bias in Long-Context LLMs **Findings of ACL 2025**

Runchu Tian*, Yanghao Li*, Yuepeng Fu, Siyang Deng, Qinyu Luo, Cheng Qian, Shuo Wang, Xin Cong, Zhong Zhang, Yesai Wu, Yankai Lin, Huadong Wang, Xiaojiang Liu

Overview: Revealed spacing-induced positional bias in long-context LLMs.

DebugBench: Evaluating Debugging Capability of Large Language Models **Findings of ACL 2024**

Runchu Tian*, Yining Ye*, Yujia Qin, Xin Cong, Yankai Lin, Yinxu Pan, Yesai Wu, Haotian Hui, Weichuan Liu, Zhiyuan Liu, Maosong Sun

Overview: Built a multi-language benchmark for evaluating large language models on code debugging.

Exploring Format Consistency for Instruction Tuning **TMLR**

Shihao Liang*, **Runchu Tian***, Kunlun Zhu*, Yujia Qin, Huadong Wang, Xin Cong, Zhiyuan Liu, Xiaojiang Liu, Maosong Sun

Overview: Studied how instruction-format consistency affects instruction tuning performance.

ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs **ICLR 2024 Spotlight**

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, **Runchu Tian**, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, Maosong Sun

Overview: Introduced ToolBench, ToolEval, and ToolLLaMA for LLM tool-use learning with real-world APIs.

Tool Learning with Foundation Models **ACM Computing Surveys**

Yujia Qin, Shengding Hu, Yankai Lin, Weize Chen, Ning Ding, Ganqu Cui, Zheni Zeng, Yufei Huang, Chaojun Xiao, Chi Han, Yi Ren Fung, Yusheng Su, Huadong Wang, Cheng Qian, **Runchu Tian**, Kunlun Zhu, Shihao Liang, et al.

Overview: Surveyed tool learning with foundation models and organized the field around unified tasks and methods.

Teaching

Teaching Assistant, CS 173 - Discrete Structures, UIUC Sept 2024 – Dec 2024

Honors

- Tsinghua University Academic Excellence Scholarship, 2021
- Tsinghua University Academic Excellence Scholarship, 2022